

Position Size and Exit Rules for Single-Leg Delta-Hedged SPY Options

Strategy research · L/S Gamma Desk, QUANTT

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Question: for the size and exit rules lets also run a /strategy-research on this

Method: Searched all 27 papers in `docs/papers/` (page-tagged index) + web search + the desk's v1 rules (`archive/v1-ls-gamma README`). Quotes verified against the cited PDF pages. Page numbers are PDF pages; printed page numbers are given where they differ.

Related research: Target DTE, moneyness, calls vs puts · Signal generation

Desk decisions already made (PM, 2026-10-01): v1 VRP band; single-leg calls and puts; daily delta hedging.

1 Short answer

The papers agree on **four kinds of exit**:

1. **Time:** get out before the final week.
2. **Signal reversal:** the thesis is gone.
3. **Loss limit:** a hard risk stop.
4. **Profit take:** optional.

For **size**, the serious frameworks either cap a risk measure (vega, max loss, collateral) or use a fraction of Kelly. Kelly needs P&L statistics the desk doesn't have yet.

For paper trading, the simplest sound start is **1 contract per trade** with time, signal and loss exits. Scale up only once real P&L data exists.

2 What the papers say

2.1 Time exits: avoid the final week

- **PRISM rolls before the last week:**

“VRC rolls all options positions when time to expiry falls below 5 trading days.” (Verma 2026, *PRISM*, p. 36)

The reason given is that it:

“avoids the theta cliff of the final week” (*PRISM*, p. 36)

- **v1 desk rule:** enter at 7–28 DTE and exit no later than 3 DTE (v1 README).
- **The Cboe PutWrite benchmark is the opposite extreme:** it sells 1-month ATM puts monthly and **holds them to expiry**, fully collateralized (Wikipedia, Cboe methodology). That index isn't delta-hedged, though.

2.2 Signal exits: close when the thesis reverses

- **PRISM’s VRP reversal trigger** fires when the VRP z-score stays negative for three consecutive closes. This:

“signals that implied vol has fallen below VRC’s regime-adjusted realized vol estimate” (*PRISM*, p. 37)

- **OPHR’s baselines close and flip:**

“If predicted volatility crosses the opposite threshold, the position is closed and reversed.” (Chen, Cai, Qin & An, *OPHR*, p. 28)

2.3 Loss and profit limits

- **OPHR’s baselines use percentage limits plus a maximum hold:**

“the position is closed if the profit reaches p% or if the loss reaches l%. In addition, a maximum holding period” (*OPHR*, p. 26)

Their settings (take-profit 5%, stop 3%, 96-hour maximum hold, 60–90 day options, p. 27) are tuned for hourly crypto trading and **don’t transfer** to daily SPY.

- **v1 desk rules:** for long gamma, a VRP hard stop and a realized-vol trailing check; for short gamma, a stop at 30% of premium.
- **Profit taking at 50% of premium** on short puts is common practitioner advice. I couldn’t verify a rigorous study (the one found couldn’t be accessed), so treat it as untested.

2.4 Sizing

- **Fractional Kelly:**

“VRC applies a fractional Kelly factor of 1/2 as standard policy” (*PRISM*, p. 23)

Kelly needs the mean and variance of trade P&L, which the desk will only have after paper trading.

- **PRISM’s hard limits (p. 38):**

- net vega at most 3% of assets
- 99% daily CVaR at most 0.75% of notional
- delta exposure at most 2× assets
- exit if ATM bid-ask spreads exceed 100bp of vol

- **The PutWrite index** sizes so that T-bills “can finance the maximum possible loss from final settlement” of the puts sold. That’s the most conservative rule.
- **v1 desk caps:** long gamma at 5–10% of capital; a 20% budget for the short side.

3 How the papers connect

Source	Exit approach	Sizing approach
PRISM (whitepaper)	Roll at under 5 trading days; signal reversal; P&L at 2.5σ below expectation; hard limits	Half-Kelly with an uncertainty penalty; vega, CVaR and leverage caps
OPHR baselines	Take-profit, stop-loss, max hold; close and reverse on opposite signal	By margin use
Cboe PutWrite (web) v1 desk	Hold to expiry Exit at 3 DTE; VRP and premium stops	Fully collateralized 5–10% long, 20% short budget

- **Everyone uses a time exit and a thesis exit.** They differ mostly on whether to add profit and loss stops.
- **Sizing goes from simplest (full collateral) to most data-hungry (Kelly).** Risk caps such as vega or max loss sit in between.

4 Relevance to the LS Gamma desk

Suggested starting rules for paper trading. These are suggestions; the PM sets the numbers.

1. **Size:** 1 contract per trade. Paper trading is first a mechanics test. Later, size by a vega or max-loss budget, and only consider half-Kelly once there's P&L history.
2. **Time exit:** close at about 7 calendar DTE, which is roughly PRISM's 5 trading days.
3. **Signal exit:** close when the signal is back to flat for 3 consecutive days. Close and flip on an opposite signal.
4. **Loss stop:**
 - short put: when the position loss reaches $1-2\times$ the premium received
 - long call: at 50% of the premium paid

Measure losses including the hedge P&L.

5. **Optional profit take:** close short puts at 50% of premium. Untested; consider running it in shadow to compare.
6. **Log every exit and its trigger**, so the rules can be tuned from data.

5 Gaps and open questions

- **No paper tests these exits on daily-hedged single SPY options.** The OPHR numbers are crypto; PRISM is simulated; PutWrite is unhedged.
- **Stops and daily hedging interact.** A stop on option P&L alone ignores hedge gains, so P&L should be measured on the combined position.
- **The account's starting capital and risk budget** aren't set yet, and they're needed before scaling past 1 contract.

6 Sources

Repo papers used, with pages cited:

- PRISM-A-Probabilistic-Regime-Integrated-Scalping-Model-for-Derivatives-Arbitrage.pdf (pp. 23, 36, 37, 38)
- OPHR-Mastering-Volatility-Trading-with-Multi-Agent-Deep-Reinforcement-Learning.pdf (pp. 26, 27, 28)

Checked, nothing on sizing or exits: the deep hedging papers (they size hedges, not positions), the forecasting papers, Agents-Are-not-Algorithms.pdf, From-Deep-Learning-to-LLMs-A-Survey-of-AI-in-Quantitative-Investment.pdf, Sepp (hedging frequency, not exits).

Web sources (outside the repo):

- Cboe S&P 500 PutWrite Indices methodology
- CBOE S&P 500 PutWrite Index (Wikipedia)
- Ennis Knupp, *Evaluating the Performance Characteristics of the CBOE S&P 500 PutWrite Index*

Suggested papers to add to docs/papers/:

1. **Ennis Knupp PUT index evaluation:** a long-run benchmark for a systematic ATM put-selling program.
2. **A rigorous study of profit and loss rules for short index options,** if one can be found. The practitioner claims remain unverified.